Automotive Technical Service Diagnostic Bulletin

Document ID: DIAG-C-CLASS-P0101
Classification: OEM Level 3 Diagnostic & Repair Procedure
Target Vehicle: Mercedes-Benz C-Class (W206 / C 300)
Powertrain: 2.0L Turbocharged Inline-4 (M254) with 2nd Gen ISG (Beltless 48V Mild-Hybrid)
Engine Control System: Bosch MD1CS006 (Engine Control Module N3/10) - High-Speed CAN-E2 Architecture
Electrical Architecture: Integrated 12V / 48V Architecture with 48V DC/DC Converter (N83/1)
Fuel Injection Architecture: Single Cam-Driven High Pressure Pump (HPP) with Solenoid Direct Injectors (200-250 bar)
Firing Order: 1-3-4-2

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1. DIAGNOSTIC TROUBLE CODE (DTC) SPECIFICATION

- **DTC Code:** `P0101`
- **Standard Fault Description:** Mass Air Flow (MAF) Sensor Circuit Range/Performance
- **System Definition:** Calculated engine load and throttle angle mismatch with hot-film air mass sensor frequency/voltage output.
- **MIL (Malfunction Indicator Lamp) Status:** Illuminated (Continuous or Flashing depending on catalyst severity)
- **Monitoring Condition:** Continuous monitoring during engine running cycle with coolant temp > 60°C and vehicle speed > 0 km/h.

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2. PLATFORM-SPECIFIC ROOT CAUSE ANALYSIS (RCA)

Root cause analysis customized specifically for the **Mercedes-Benz C-Class (W206 / C 300)**:

1. **Hot-Wire / Film Sensor Contamination from Engine Air Filter Fitment:** Aftermarket air filter with poor perimeter gasket letting fine dust bypass onto sensing film.
2. **Cold Charge Air Pipe Resonator Crack:** Plastic weld split along charge air duct between turbo compressor outlet and intercooler.
3. **Throttle Body M16/6 Carbon Edge Build-up:** Inducing abnormal manifold depression that fails the mass air flow plausibility calculation model in MD1 ECM.

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3. DEDICATED WIRING DIAGRAM & CIRCUIT PINOUT

Schematic configuration and electrical pin connections specifically for the C-Class control harness:

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[Digital Mass Air Flow Sensor B2/5]        [ECM N3/10 Connector 'K']
Pin 1 (IAT Signal 0-5V) ------------------> Pin 37 (Intake Temp Signal)
Pin 2 (+12V Battery Run/Start) <---------- Fuse F15 (Front SAM N10/1)
Pin 3 (Chassis Ground) -------------------> Ground Stud W9 (Wheel Arch)
Pin 4 (+5V Reference Output) <------------ Pin 38 (Regulated 5V)
Pin 5 (PWM Frequency Airflow Signal) ----> Pin 39 (MAF Frequency In)
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- Airflow Spec: 8.5 kg/h - 12.0 kg/h at curb idle (600 RPM, AC Off).

Electrical Reference Points

- **Primary Ground Reference:** `W10 (Engine Block Ground Point, Right) & W9 (Front Wheel Arch Ground)`
- **Power Distribution Node:** `Engine Bay Prefuse Box F32/1 & Front SAM Module (N10/1)`
- **Signal Logic Voltage Range:** 0.0V to 5.0V regulated DC (unless high-voltage direct injection rail)

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4. STEP-BY-STEP DIAGNOSTIC & VERIFICATION PROCEDURE

Follow this OEM workshop sequence to isolate the failure on this platform:

1. Check MAF reading in XENTRY: compare mass flow against nominal calculated mass flow window.
2. Perform smoke test on charge air pipe between turbocharger, charge air cooler, and intake manifold.
3. Clean throttle body throat of carbon deposits using lint-free cloth and solvent.
4. Verify air filter sealing gasket seating inside air box.

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5. PRECISE RESOLUTION & REPAIR PROCEDURE

- Replace Mass Air Flow Sensor B2/5 with genuine OEM part.
- Install new genuine Mercedes-Benz engine air filter (Part # A254094xxxx).
- Perform adaptation reset: "Engine -> Adaptations -> Reset Air Flow Calculation Matrix".

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6. POST-REPAIR QUALITY VALIDATION & RE-TESTING

1. Clear all DTCs from Engine Control Module (`Bosch`) and secondary gateway modules.
2. Perform comprehensive drive cycle:
3. Cold start idle for 3 minutes (monitors secondary air / catalyst preheat and cold misfire).
4. Accelerate smoothly to 50 km/h and hold for 2 minutes.
5. Cruise at 80–100 km/h on highway for 10 minutes (validates downstream O2 sensor and fuel trim closed-loop balance).
6. Coasting deceleration (fuel cut-off check) for 5 seconds without braking.
7. Re-scan control units with diagnostic tool; confirm all OBD-II readiness monitors show **READY** / **COMPLETE** and no pending DTCs are stored.